

## Peer Review File

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### Review Comments

The paper titled “Identification of the Key Gene for Hepatocellular Carcinoma Based on Bioinformatics and Machine Learnings and Experimental Verification” is interesting. FAM83D may be a key gene involved in the development and progression of HCC, contributing to early diagnosis and prognosis assessment. It has the potential to serve as a biomarker for HCC. However, there are several minor issues that if addressed would significantly improve the manuscript.

1) The statement that "It is still unclear how aberrant gene expression contributes to HCC carcinogenesis" is overly absolute and inconsistent with the numerous existing studies on the molecular mechanisms of HCC.

**Reply:** Thank you for pointing out the shortcomings in our manuscript. We acknowledge that our previous wording was overly absolute and have now revised this section accordingly.

**Changes in the text:** In page 1, line 20-24. “While deregulated gene expression has been widely linked to hepatocarcinogenesis, many details of how these alterations drive tumor initiation and progression remain to be elucidated. We therefore combined bioinformatics and machine-learning strategies to screen for and extensively validate candidate driver genes in HCC.”

2) The core objective has not been clearly defined. It is suggested to clarify whether the genes being screened are diagnosis-related, prognosis-related, or those with both diagnostic and prognostic value, and to mention whether the functional mechanisms of the key genes have been explored.

**Reply:** Thank you for your meticulous review. Our study first screened genes closely associated with HCC diagnosis and then identified those among them that significantly influence patient prognosis. The final candidate, FAM83D, is not only strongly linked to HCC diagnosis but also provides valuable guidance for predicting patient prognosis.

**Changes in the text:** None.

3) The association analysis of immune cells was not in-depth. It is suggested to explain the specific association direction and correlation coefficient between FAM83D expression and various immune cells, and to explain the impact of this association on the immune microenvironment of HCC.

**Reply:** Thank you for your careful review. In response to your suggestion, we have now included the correlation coefficients between FAM83D expression and the abundance of infiltrating immune cells, enabling readers to better appreciate the potential regulatory link between FAM83D levels and the immune microenvironment.

**Changes in the text:** In page 11, line 278-285. “FAM83D had a significant positive correlation with immunosuppressive cells, such as T cells regulatory (Tregs,  $R=0.34$ ,  $P<0.001$ ), M2 macrophages ( $R=0.28$ ,  $P=0.002$ ), resting dendritic cells, resting mast

cells ( $R=0.24$ ,  $P=0.011$ ), and M0 macrophages ( $R=0.20$ ,  $P=0.030$ ), and a significant negative relation to immune-stimulating cells, like gamma delta T cells ( $R=-0.20$ ,  $P=0.029$ ), activated mast cells ( $R=-0.21$ ,  $P=0.024$ ), plasma cells ( $R=-0.22$ ,  $P=0.016$ ), B cells naive ( $R=-0.23$ ,  $P=0.013$ ), and CD4+ T cells naive ( $R=-0.37$ ,  $P<0.001$ ) (Fig. 9B-M)”

4) Based solely on the expression differences and prognostic associations, without verifying its functions through cell experiments, the conclusion remains at the correlation level. It is necessary to weaken the expression or supplement functional experimental hints.

**Reply:** Thank you for your insightful comments. We have now validated the findings in an independent cohort of 100 cases by immunohistochemistry (IHC), confirming that protein expression is significantly associated with prognosis and strengthening our conclusions at the in-situ tissue level. Because re-establishing stable cell lines for functional assays would require extensive laboratory resources and new biosafety approvals, we cannot complete these experiments in the short term; this limitation has been added to the Discussion.

**Changes in the text:** page15, line 395-396: “In addition, we lack further functional experiments to prove how FAM83D affects the occurrence and development of HCC.”

5) It is recommended to provide evidence of clinical applicability, such as comparing the diagnostic efficacy of FAM83D with existing markers, and explaining the expression differences in early HCC.

**Reply:** We appreciate your suggestion and have incorporated the new GSE63898 dataset, which contains sequencing data from early-stage HCC patients, to further validate the diagnostic performance of FAM83D for early-stage HCC.

**Changes in the text:** In page 6, line 152-154. “The GSE63898 dataset, (including 228 HCC and 168 nHCC samples), was used to validate the performance of the FAM83D gene in diagnosing early-stage HCC. Early-stage HCC was defined as BCLC stage 0-A.”

6) No analysis results from single-cell and spatial transcriptomics were mentioned in the results section. If these advanced analyses are conducted, the results must be reflected in the abstract.

**Reply:** Thank you for highlighting the incompleteness in our abstract. We have now added the corresponding findings from both single-cell and spatial transcriptomic analyses to the results section of the abstract, ensuring that the methods and results are fully aligned.

**Changes in the text:** In page 2, line 54-56. “Further single-cell analysis revealed that FAM83D is predominantly upregulated in malignant cells, and its high expression is strongly associated with poor response to immunotherapy in HCC patients”

7) Jumping directly to "Confirming FAM83D as the key gene" lacks a convincing logical bridge in the middle. Readers don't know how FAM83D stood out among 80 DEGs? Please explain how many gene intersections were screened out by the two

machine learning algorithms, what the ranking of FAM83D is in this intersection, and what its diagnostic efficacy is in the training set and validation set.

**Reply:** Thank you for this constructive suggestion. We have now added a complete description of the filtering pipeline in Results section 2.4 and Figs. 4A–B, 7A–G and 8. Briefly, LASSO and SVM-RFE were applied to the 80 DEGs, yielding an intersecting set of seven genes (CXCL14, EPCAM, HSD17B2, LY6E, FAM83D, CDHR2 and SMIM24). ROC analysis in the test cohort identified four genes with  $AUC > 0.9$ , of which only FAM83D showed significant prognostic value (fig.8,  $P < 0.05$ ); thus, FAM83D was advanced as the key gene for all downstream analyses.

**Changes in the text:** None.